

SMART LENDER

ML-Powered Loan Approval Prediction System

Project Documentation | SmartBridge Internship | 2026

Detail	Information
Project Name	Smart Lender
Domain	Finance / FinTech / Machine Learning
Developer	Lahari
Institute	Srinivasa Ramanujan Institute of Technology
Platform	SmartBridge
Year	2026
Best Model	XGBoost – 94.7% Train 81.1% Test Accuracy
Deployment	Render (Free Tier)

1. Abstract

Smart Lender is a machine learning-powered web application designed to predict the creditworthiness of loan applicants, enabling banks and financial institutions to make faster, data-driven loan approval decisions. The platform leverages four classification algorithms — Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost — to evaluate applicant data and determine the likelihood of loan repayment or default.

After training and evaluating all four models, the best-performing model (XGBoost at 94.7% training accuracy and 81.1% testing accuracy) is saved and integrated into a Flask web application for real-time prediction. The system empowers financial analysts, credit officers, and banking professionals to reduce non-performing assets and improve the overall credit evaluation process with confidence.

2. Problem Statement

Traditional loan approval processes are slow, inconsistent, and often dependent on subjective human judgment. Financial institutions face significant challenges in:

- Identifying high-risk applicants accurately before disbursement
- Handling large volumes of loan applications during peak periods
- Reducing non-performing assets (NPAs) caused by defaults
- Ensuring compliance and fairness in lending decisions

Smart Lender automates and standardizes this process using machine learning, enabling accurate, fast, and unbiased credit evaluations at scale.

3. Features

- Trains and compares 4 ML models: Decision Tree, Random Forest, KNN, XGBoost
- Automated best-model selection based on test accuracy
- EDA visualizations: income distribution, credit history analysis, model comparison
- Feature importance chart for model interpretability
- Confusion matrix for performance evaluation
- Real-time prediction via Flask web application
- Clean, responsive dark-themed UI
- One-click deployment on Render (free tier)

4. Tech Stack

Layer	Technology	Purpose
Language	Python 3.8+	Core development language
Web Framework	Flask 2.x	REST API & web interface
ML – Ensemble	XGBoost, Random Forest	Best-performing models
ML – Baseline	Decision Tree, KNN	Comparison baselines
ML Library	Scikit-Learn	Preprocessing, metrics
Data Processing	Pandas, NumPy, SciPy	Data cleaning & analysis

Visualization	Matplotlib, Seaborn	EDA & performance plots
Model Saving	Joblib	Pickle best model
Deployment	Render / IBM Cloud	Live demo hosting

5. Model Performance

Model	Train Accuracy	Test Accuracy	Status
Decision Tree	~100%	~71%	Overfits
Random Forest	~100%	~78%	Good
KNN	~83%	~72%	Moderate
XGBoost ★	94.7%	81.1%	Best – Saved

XGBoost was selected as the best model due to its superior test accuracy and balanced bias-variance trade-off. It was saved as `best_model.pkl` and integrated into the Flask application.

6. Dataset

Attribute	Detail
Source	Kaggle – Loan Prediction Problem Dataset
Records	614 rows
Features	12 input features + 1 target (Loan_Status)
Target	Loan_Status: Y (Approved) / N (Rejected)
Link	kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset

Input Features

Feature	Type	Description
Gender	Categorical	Male / Female
Married	Categorical	Yes / No
Dependents	Categorical	0, 1, 2, 3+
Education	Categorical	Graduate / Not Graduate
Self_Employed	Categorical	Yes / No
ApplicantIncome	Numerical	Monthly income of applicant
CoapplicantIncome	Numerical	Monthly income of co-applicant
LoanAmount	Numerical	Loan amount in thousands
Loan_Amount_Term	Numerical	Term in months
Credit_History	Binary	1 = Good, 0 = Bad
Property_Area	Categorical	Urban / Semiurban / Rural

7. Installation Steps

Step 1 – Clone the Repository

```
git clone https://github.com/YOUR_USERNAME/smart-lender.git
cd smart-lender
```

Step 2 – Install Dependencies

```
pip install -r requirements.txt
```

Step 3 – Download Dataset

Download loan_data.csv from Kaggle and place it in the project root folder.

Step 4 – Train the Model

```
python train_model.py
```

Step 5 – Verify Model

```
python verify_model.py
```

Step 6 – Run the Flask App

```
python app.py
```

Open browser at: <http://127.0.0.1:5000>

8. Usage Instructions

- Open the web application in your browser
- Fill in the applicant's personal and financial details in the form
- Click the 'Predict Loan Approval' button
- The system instantly displays: ✓Loan Approved or ✗Loan Rejected
- Credit officers can use this to fast-track low-risk applicants
- Analysts can batch-evaluate applicants during high-volume periods

9. Real-World Scenarios

Scenario 1 – Fast-Track Approval

A bank credit officer enters details of a salaried applicant with good credit history and stable income. The model predicts loan approval with high confidence, allowing fast-tracking without manual review.

Scenario 2 – High-Risk Detection

An applicant with irregular self-employment income and no credit history submits a request. The system flags it as high-risk, prompting further scrutiny or document verification before proceeding.

Scenario 3 – Batch Evaluation

A financial analyst uses the platform to evaluate multiple applicants during peak periods. The XGBoost model significantly reduces evaluation time while maintaining accuracy and compliance.

10. Future Scope

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- Add probability confidence scores alongside binary predictions
 - Integrate SHAP values for model explainability and transparency
 - Add user authentication system for credit officers and analysts
 - Connect to live banking database via secure REST API
 - Extend to multi-class risk scoring: Low / Medium / High risk
 - Build mobile application version using React Native
 - Add batch CSV upload for processing multiple applicants at once
 - Implement model retraining pipeline with new data

11. Team Member Details

Name	Role	Responsibilities
Lahari Gadamsetty (Team Leader)	Team Leader	Data preprocessing, model training, Flask app, deployment
Uppara Anusha	Team Member	ML model development, data analysis
Praneetha Bandi	Team Member	Data preprocessing, EDA, visualizations
Juturu Vyshnavi	Team Member	Flask UI development, testing
Sathvika Godena	Team Member	Documentation, project reporting